observed that below $400 it was one of the most undervalued large capitalization stock in the United States, and, as we know, undervaluation is a condition for future market-beating returns.

Shortly after Graham published *Security Analysis* in 1934, John Burr Williams published his 1938 masterpiece *The Theory of Investment Value*.[44] Williams's *discounted cash flow* theory of intrinsic value is the bedrock of modern valuation and forms the intellectual basis for a variety of valuation models, including, for example, Gordon's growth model, which we saw earlier. The models all require an estimate for future cash flows, earnings, or dividends, making allowance for any growth, and out to perpetuity. Money has a time value—a dollar today is worth more than a dollar in a year—so we must then discount back to today those future cash flows at the appropriate discount rate. While the theory is sound, the slip 'twixt cup and lip is in its practical application. The three variables—future cash flows, the growth rate, and the discount rate—are all sources of potential forecast error. Discounted cash flow models are extremely sensitive to discount rates, which can lead to large errors, but the real problem is that the model assumes we have some way of forecasting future cash flows and the growth rates embedded in them. Time and again, we have been shown to be poor forecasters, preferring to extrapolate the current trend, rather than assume mean reversion. Kahneman and Tversky call this the "misconception of regression," and it rears its head time and again in investment.

Studies of financial analysts' earnings forecasts, for example, have found them to fail to incorporate into the forecast an expectation of mean reversion when it is bound to occur, and therefore to deliver results that are *no better than random chance*. In 2007 Roy Batchelor examined the records of financial analysts over the period 1990 to 2005, and found them to be consistently wrong, persistently erring on the side of over-optimism because they extrapolated up an existing trend without allowing for mean reversion. Figure 6.2 is a chart showing Batchelor's finding that analysts tend to be too optimistic.[45]

The chart shows that forward earnings forecasts are rarely correct and tend to overshoot actual earnings. Batchelor finds this bias to over-optimism to be systematic. The size of the forecast error declines when economic growth accelerates and increases when economic growth slows. In Batchelor's research, actual earnings beat the forecasts only twice in 25 years—both times during recoveries following a recession. When the forecasts did hit the target it was because the underlying economic growth was unusually strong, which is to say the forecast turned out to be right because the usual assumptions in the forecast were wrong. The analysts missed completely any turn in the fortunes of the market or the economy. Like naïve extrapolation investors, they failed to account for any possibility of mean reversion when mean reversion is the *probable* outcome.